Lab 15 – Debugging Report: Network Errors in NewsReader

Mobile Application Programming (71ITSE30603 / 72ITSE30603) • Student ID: _____ • Full name: _____ • Device: API 34 emulator

Error 1 — android.os.NetworkOnMainThreadException

How reproduced: set `DebugChallenge.TRIGGER_NETWORK_ON_MAIN_THREAD = true`, which calls `HttpClient.getBlocking()` directly inside `onCreate()` without `withContext(Dispatchers.IO)`. Logcat filter: `tag:DEBUG_CHALLENGE`.

```
E DEBUG_CHALLENGE: Reproduced NetworkOnMainThreadException
    android.os.NetworkOnMainThreadException
        at android.os.StrictMode$AndroidBlockGuardPolicy.onNetwork(StrictMode.java:1694)
        at java.net.Inet6AddressImpl.lookupHostByName(Inet6AddressImpl.java:115)
        at java.net.Inet6AddressImpl.lookupAllHostAddr(Inet6AddressImpl.java:103)
        at java.net.InetAddress.getAllByName(InetAddress.java:1152)
        at com.android.okhttp.Dns$1.lookup(Dns.java:41)
        ...
        at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getResponseCode(HttpURLConnectionImpl.java:542)
        at com.android.okhttp.internal.huc.DelegatingHttpsURLConnection.getResponseCode(DelegatingHttpsURLConnection.java:106)
        at com.vlu.newsreader.network.HttpClient.executeGet(HttpClient.kt:38)
        at com.vlu.newsreader.network.HttpClient.getBlocking(HttpClient.kt:28)
        at com.vlu.newsreader.debug.DebugChallenge.runNetworkOnMainThread(DebugChallenge.kt:32)
        at com.vlu.newsreader.MainActivity.onCreate(MainActivity.kt:55)
        at android.app.Activity.performCreate(Activity.java:8595)
        ...
        at android.os.Looper.loop(Looper.java:317)
        at android.app.ActivityThread.main(ActivityThread.java:8705)
```

Root cause: Android runs all UI drawing and input on one thread (the main thread). A network call can take seconds; if it ran there, the screen would freeze and the system would show an "App Not Responding" dialog after about 5 s. Since API 11, StrictMode blocks this: the first time the request touches the network (here the DNS lookup made when `getResponseCode()` opens the connection) it throws this exception. The key line is `HttpClient.kt:38` being called from `MainActivity.onCreate`.
Fix: run the request inside `lifecycleScope.launch { }` and wrap it in `withContext(Dispatchers.IO)` (`HttpClient.get()`, line 21). The coroutine suspends instead of blocking, the main thread stays free, and the result comes back to Main to update the RecyclerView.

Error 2 — java.lang.SecurityException (missing INTERNET permission)

How reproduced: commented out `<uses-permission android:name="android.permission.INTERNET"/>`, uninstalled the app (so the old grant is removed), and ran again. The app did not crash: the `catch` block logged it under tag `NewsReader` and showed the "missing INTERNET permission" message.

```
E NewsReader: Failed to load news
    java.lang.SecurityException: Permission denied (missing INTERNET permission?)
        at java.net.Inet6AddressImpl.lookupHostByName(Inet6AddressImpl.java:150)
        at java.net.Inet6AddressImpl.lookupAllHostAddr(Inet6AddressImpl.java:103)
        at java.net.InetAddress.getAllByName(InetAddress.java:1152)
        ...
        at com.android.okhttp.internal.huc.DelegatingHttpsURLConnection.getResponseCode(DelegatingHttpsURLConnection.java:106)
        at com.vlu.newsreader.network.HttpClient.executeGet(HttpClient.kt:38)
        at com.vlu.newsreader.network.HttpClient.access$executeGet(HttpClient.kt:13)
        at com.vlu.newsreader.network.HttpClient$get$2.invokeSuspend(HttpClient.kt:21)
        at kotlin.coroutines.jvm.internal.BaseContinuationImpl.resumeWith(ContinuationImpl.kt:33)
        at kotlinx.coroutines.DispatchedTask.run(DispatchedTask.kt:101)
        at kotlinx.coroutines.scheduling.CoroutineScheduler$Worker.run(CoroutineScheduler.kt:684)
    Caused by: android.system.GaiException: android_getaddrinfo failed: EAI_NODATA (No address associated with hostname)
        at libcore.io.Linux.android_getaddrinfo(Native Method)
        ...
    Caused by: android_getaddrinfo failed: EAI_NODATA
```

Root cause: Every Android app runs in a sandbox. An app gets network sockets only if it declares `INTERNET` in the manifest (a "normal" permission, granted automatically at install). Without it the app's Linux user is not in the network group, so the kernel refuses the DNS lookup (`EAI_NODATA` / `EPERM`), and the runtime turns that into a `SecurityException`. Notice the trace runs on a `CoroutineScheduler$Worker` thread, which shows the request was correctly on `Dispatchers.IO`. The thread was fine; the permission was missing.
Fix: put the `<uses-permission>` line back as a direct child of `<manifest>` (outside `<application>`) and reinstall.

Comparison and lessons

	Error 1: NetworkOnMainThread	Error 2: SecurityException
Kind of mistake	Threading / architecture	Configuration (manifest)
Thread in trace	main (ActivityThread.main)	DefaultDispatcher worker (IO)
Effect if not caught	App crashes (FATAL EXCEPTION: main)	Crashes the coroutine; caught → error UI
Fix	withContext(Dispatchers.IO)	Add INTERNET uses-permission

Reading a trace: start at the exception name, then scan down for the first line in `com.vlu.newsreader`; that is where my code caused it. Check the bottom frames to see which thread it ran on. Airplane Mode gives a different error (`UnknownHostException`), which the app shows as "No internet connection" with a Retry button.